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Affect analysis in context of characters in narratives

Michal Ptaszynski [a,*], Hiroaki Dokoshi [b], Satoshi Oyama [b], Rafal Rzepka [b], Masahito Kurihara [b], Kenji Araki [b], Yoshio Momouchi [c]

[a] Hokkai-Gakuen University, High-Tech Research Center, Minami 26, Nishi 11, Chuo-ku, Sapporo 064-0926, Japan
[b] Hokkaido University, Graduate School of Information Science and Technology, Kita-ku Kita 14 Nishi 9, 060-0814 Sapporo, Japan
[c] Hokkai-Gakuen University, Department of Electronics and Information Engineering, Faculty of Engineering, Minami 26, Nishi 11, Chuo-ku, Sapporo 064-0926, Japan

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ABSTRACT

This paper presents our research in text-based affect analysis (AA) of narratives. AA represents a task of estimating or recognizing emotions elicited by a certain semiotic modality. In text-based AA the modality in focus is the textual representation of language. In this research we study particularly one type of language realization, namely narratives (e.g., stories, fairy tales, etc.). Affect analysis within the context of narratives is a challenging task because narratives are created of different kinds of sentences (descriptions, dialogs, etc.). Moreover, different characters become subjects of different emotional expressions in different parts of narratives. In this research we address the problem of person/character related affect recognition in narratives. We propose a method for emotion subject extraction from a sentence based on analysis of anaphoric expressions and compare two methods for affect analysis. We evaluate the system and discuss its possible future improvements.

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1. Introduction

Understanding emotions expressed in language is an important part of communication between humans. Along with the development of language and writing people were able to develop various strategies for expressing emotions through text. Although one of the most obvious situations in which we express emotions is a conversation, the emotions have been expressed in many different types of language vehicles. Some of the examples include epistolary art (e.g., love letters), personal diaries (with their more recent version – blogs) or stories (e.g., fairy tales). These all are part of one category, namely narratives. What differentiates narratives from conversation is its descriptive force. While in conversations most of the time the subject is the first person (“me/I”), narratives provide an opportunity to describe a situation through the eyes of a third person. Especially in children stories and fairy tales this becomes an apparatus to teach the readers (children) important emotional strategies. For example, the fear elicited toward Little Red Riding Hood being devoured by the Bad Wolf helps children understand that they should not talk to strangers. Or when the little Goat Child, from the story of “The Wolf and the Seven Young Kids”, is saved from being eaten and runs to get help, children can learn that in crisis situations it is the best to run and look for help in adults. Finally, when the heroes of such stories are rescued in the end the children can understand the value of relief and happiness after a difficult struggle.

Understanding emotions in narratives, and especially in children stories, is therefore a crucial task to develop a computational model of language understanding. In this paper we address the task of recognition of emotions (later called affect analysis) in children stories. In general, affect analysis within the context of narratives is a challenging task for several reasons. Firstly, because narratives are created of different kinds of sentences, such as descriptive sentences (pushing the story forward) or dialogs (e.g., conversations between story characters). Secondly, because different characters become subjects of different emotional expressions in different parts of narratives. In this research we especially focused on this particular problem, namely person/character related affect recognition in narratives. At first we developed a method for extraction of emotion subjects from sentences. The method is based on analysis of anaphoric expressions. Next, the affect analysis procedure is performed separately for all recognized characters in each part of the story. We compared two versions of the affect analysis procedure. The first version was based on Japanese version of WordNet, a thesaurus consisting of words with specified relations to each other. The second method was based on ML-Ask, a system for affect analysis created especially for Japanese language. Finally, we evaluated the method on the story “The Wolf and the Seven Young Kids” by the Grimm Brothers.

The outline of this paper is as follows. In Section 2 we present the background for this study by describing other research on affect analysis in general, in narratives and in Japanese language.

* Corresponding author.
E-mail address: ptaszynski@hgu.jp (M. Ptaszynski).

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In Section 3 we describe the previously developed tools we apply in our methods. In Section 4 we present the method allowing simultaneous affect analysis of separate characters in narrative. We evaluate the method in Section 5. Finally, in Section 6, we conclude the paper and propose some ideas to improve the described method as well as its possible future applications.

2. Affect analysis in narratives

Text based affect analysis (AA) has been defined as a field focused on developing natural language processing techniques for estimating the emotive aspect of text (Grefenstette, Qu, Shanahan, & Evans, 2004). For example, Elliott (1992) proposed a keyword-based Affect analysis system applying an affect lexicon (including words like "happy", or "sad") with modifiers ("extremely", "somewhat"). Liu, Lieberman, and Selker (2003) presented a model of text-based affect sensing based on OMCS (Open-Mind Common Sense), a generic common sense database, with an application to e-mail interpretation. Alm, Roth, and Sproat (2005) proposed a machine learning method for affect analysis of fairy tales. Aman and Szpakowicz also applied machine learning techniques to analyze emotions expressed on blogs (Aman & Szpakowicz, 2007).

There have also been several attempts to achieve this goal for the Japanese language. For example, Tsuchiya, Yoshimura, Watabe, and Kawaoka (2007) tried to estimate emotive aspect of utterances with the use of an association mechanism. On the other hand, Tokuhisa, Inui, and Matsumoto (2008) as well as Shi, Rzepka, and Araki (2008) used a large number of examples gathered from the Web to estimate user emotions. Furthermore, Ptaszynski, Dybala, Shi, Rzepka, and Araki (2009a)a proposed a Web-based supported affect analysis system for Japanese text-based utterances.

There have been also some research in affect analysis of narratives. For example, Alm et al. (2005) used a machine learning algorithm with multiple linguistic features to distinguish general affect expressed in stories, especially in fairy tales. Neviarouskaya, Prendinger, and Ishizuka (2010) tested their affect analysis model for English on various kinds of data, including fairy tales (the same stories as Alm et al.). Finally, Mohammad (2011) proposed a visualization method for particular emotion related words in stories based on the idea of emotion word density. However, they did not aim at precise affect detection, but rather visualization of presence of emotion words within certain children's stories (192 Brothers Grimm fairy tales) The only research for affect analysis of fairy tales for Japanese language was done by Fukkoshi and Sugimoto (2010). They used similarity score to expand a small initial set of emotion words (503 nouns) into an emotion lexicon (15,612 verbs) and performed syntactic rule matching on one fairy tale ("Snow White"). Unfortunately, their results were less than ideal due to lacks in the lexicon and inconsistencies in syntactic rules. The previous research for affect analysis of narratives described above is summarized in Table 1.

In all of the above research on affect analysis in narratives the analysis was simplified to detecting general affect for a story, or part of the story (sentence/paragraph). In this research we focused not on general affect, but tried to specify who is the subject of the particular emotion.

Previously there have been a few research on affect analysis of emotion subjects (characters/actors), both for English language. Zhang (2010) proposed affect analysis method for text-based conversations, in which she used predetermined relationship profiles. The final emotion score calculated for user utterance was influenced by the type of relationship the user had with previous speaker. For example, if user A was in a bad relationship with previous speaker B it was more likely A would express negative emotion, if B expressed positive emotion, like joy, or happiness. Another research was done by Calix and Knapp (2011). They tested linear vs. non-linear Support Vector Regression (SVR) with different kernel functions to measure actor level emotion magnitude. However, they focused mostly on variations of speech signals and simplified textual features to keyword spotting.

Comparing to all previous research, the research presented here is pioneering in several ways. Firstly, it is the first attempt to combine the tasks of text-based affect detection in fairy tales and in characters (emotion subjects). It is also the first attempt to apply thesaurus-based methods to each of these tasks and compare thesaurus-based methods (based on Japanese WordNet) to corpus-based methods (based on ML-Ask system). Finally, it is the first attempt to perform all of the above in Japanese language.

3. Tools

In this section we describe all tools and methods for basic affect analysis (AA) used further in contextual affect analysis tasks.

3.1. Emotive expression dictionary

Emotive Expression Dictionary (Nakamura, 1993) is a dictionary developed by Akira Nakamura in a period of over 20-year time. It is a collection of over two thousand expressions describing emotional states collected manually from a wide range of literature. It is not a tool *per se*, but was converted into an emotive expression database by Ptaszynski, Dybala, Rzepka, and Araki (2009a) and Ptaszynski et al. (2009a) in their research on affect analysis of utterances in Japanese. In our task we needed to choose a lexicon proved to be the most appropriate for the Japanese language. Nakamura's dictionary is a state-of-the art example of a

Table 1
Previous research on affect analysis of narratives.

	Emotion categories, dimensions	Language	Approach	Domains	Level	Dataset	Features	Techniques
Alm et al. (2005)	Anger/Disgust/Fear/Happiness/ Sadness/PositivelySurprised/ NegativelySurprised	English	Fusion-based	Fairy tales	Sentence	Children stories (185 tales)	Syntactic, Semantic, Stylistic	Winnow
Neviarouskaya et al. (2010)	Anger/Fear/Happiness/ Sadness/Surprize/NoEmotion, Valence	English	Corpus-based	Fairy tales, News, Blogs	Sentence, Document	Children stories (185 tales), News headlines (1890 sentences), Blog entries (700 sentences)	Syntactic	Affect analysis model
Fukkoshi and Sugimoto (2010)	Anger/Disgust/Fear/Happiness/ Sadness/Surprize/No Emotion, Valence	Japanese	Rule-based	Fairy tales	Sentence	1 story ("Snow White")	Syntactic	Similarity score
Mohammad (2011)	Angry/Disgust/Fear/Happiness/ Sadness/Surprise/Trust/ Anticipation, Valence	English	Corpus-based	Fairy tales	Word, Document	192 stories (The Brothers Grimm fairy tales)	Stylistic	Keyword matching

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Table 2
Distribution of separate expressions across emotion classes in Nakamura's dictionary, ordered by the number of expressions per class.

Emotion class	Number of expressions	Emotion class	Number of expressions
Dislike	532	Fondness	197
Excitement	269	Fear	147
Sadness	232	Surprise	129
Joy	224	Relief	106
Anger	199	Shame	65
		Sum	2100

hand-crafted lexicon of emotive expressions. It also proposes a classification of emotions that reflects the Japanese language and culture the most appropriately. This classification is also applied in the lexicon itself. All expressions are classified as representing a specific emotion type, one or more if applicable. In particular, Nakamura proposes ten emotion types the most appropriate for the Japanese language and culture. These are: 喜 *ki/yorokobi* (joy, delight; later referred to as **joy**), 怒 *dō/ikari* (**anger**), 哀 *ai/aware* (sorrow, sadness, gloom; later referred to as **sadness**), 怖 *fu/kowagari* (**fear**), 恥 *chi/haji* (shame, shyness, bashfulness; later referred to as **shame**), 好 *kō/suki* (liking, fondness; later referred to as **fondness**), 厭 *en/iya* (dislike, detestation; later referred to as **dislike**), 昂 *kō/takaburi* (**excitement**), 安 *an/yasuragi* (**relief**), and 驚 *kyō/odoroki*(surprise, amazement; later referred to as **surprize**). The distribution of separate expressions across all emotion classes is represented in Table 2.

A frequent manner in text-based affect analysis research is applying a list of emotion classes based on other modalities than linguistic, such as face recognition, or simply creating a new class list for the need of a particular research see for example comparison of emotion class standards in Ptaszynski, Rzepka, Araki, and Momouchi (2012). In our research we aimed at contributing to the standardization of emotion class list in the language based research on emotions in Japanese. Therefore we selected Nakamura's emotion type list, developed for over 20 years, as the most appropriate for our research.

3.2. ML-Ask

ML-Ask, or *e***M***otive e***L***ement and Expression* **A***nalysis* **s***ystem* is a keyword-based language-dependent system for automatic affect annotation on utterances in Japanese constructed by Ptaszynski et al. (2009). It uses a two-step procedure:

1. Specifying whether an utterance is emotive, and
2. Recognizing the particular emotion types in utterances described as emotive.

ML-Ask is based on the idea of two-part classification of realizations of emotions in language into:

(1) *Emotive elements* or *emotemes*, which indicate that a sentence is emotive, but do not detail what specific emotions have been expressed. For example, interjections such as "whoa!" or "Oh!" indicate that the speaker (producer of the utterance) has conveyed some emotions. However, it is not possible, basing only on the analysis of those words, to estimate precisely what kind of emotion the speaker conveyed. Ptaszynski et al. include in emotemes such groups as interjections, mimetic expressions, vulgar language and emotive markers. The examples in Japanese are respectively: *sugee* (great! – interjection), *wakuwaku* (heart pounding – mimetic), – *yagaru* (syntactic morpheme used in verb vulgarization) and '!', or '??' (sentence markers indicating emotiveness). Ptaszynski et al. collected and hand-crafted a database of 907 emotemes. A set of features similar to what is defined by Ptaszynski et al. as emotemes has been also applied in other research to discriminate between emotive (emotional/subjective) and non-emotive (neutral/objective) sentences (Wilson & Wiebe, 2005; Wiebe, Wilson, & Cardie, 2005; Aman & Szpakowicz, 2007).

(2) *Emotive expressions* are words or phrases that directly describe emotional states, but could be used to both express one's emotions and describe the emotion without emotional engagement. This group could be realized by such words as *aijou* (love – noun), *kanashimu* (to feel sad, to grieve – verb), *ureshii* (happy – adjective), or phrases such as: *mushizu ga hashiru* (to give one the creeps [of hate]) or *ashi ga chi ni tsukanai* (walk on air [of happiness]). As the collection of emotive expressions ML-Ask uses the database created on the basis of Nakamura's *Emotive Expression Dictionary* (Nakamura, 1993).

With these settings ML-Ask was proved to distinguish emotive sentences from non-emotive with a very high accuracy (over 90%) and to annotate affect on utterances with a sufficiently high Precision (85.7% compared to human annotators), and satisfying, although not ideal Recall (54.7%) (Ptaszynski et al., 2009a; Ptaszynski et al., 2009a). To improve the system performance we also implemented Contextual Valence Shifters.

The idea of Contextual Valence Shifters (CVS) was first proposed by Polanyi and Zaenen (2006). They distinguished two kinds of CVS: negations and intensifiers. The group of negations contains words and phrases like "not", "never", and "not quite", which change the valence polarity of the semantic orientation of an evaluative word they are attached to. The group of intensifiers contains words like "very", "very much", and "deeply", which intensify the semantic orientation of an evaluative word. ML-Ask fully incorporates the negation type of CVS with a 108 syntactic negation structures. Examples of CVS negations in Japanese are structures such as: *amari -nai* (not quite-), *-to wa ienai* (cannot say it is-), or *-te wa ikenai* (cannot [verb]-). As for intensifiers, although ML-Ask does not include them as a separate database, most Japanese intensifiers are included in the emoteme database. The system also calculates emotive value, or emotional intensity of a sentence, on the basis of the number of emotemes in the sentence. The performance of setting the emotive value was evaluated on 84% (Ptaszynski et al., 2009a). Finally, the last distinguishable feature of ML-Ask is implementation of Russell's two dimensional affect space (Russell, 1980). It assumes that all emotions can be represented in two dimensions: the emotion's valence or polarity (positive/negative) and activation (activated/deactivated). An example of negative-activated emotion could be "anger"; a positive-deactivated emotion is, e.g., "relief". The mapping of Nakamura's emotion types on Russell's two dimensions proposed by Ptaszynski et al. (2009a) was proved reliable in several research (Ptaszynski et al., 2009a; Ptaszynski, Dybala, Shi, Rzepka, & Araki, 2009b; Ptaszynski, Maciejewski, Dybala, Rzepka, & Araki, 2010). The mapping is represented in Fig. 1. An example of ML-Ask output is represented in Fig. 2.

3.3. Japanese WordNet

WordNet[1] (Miller, Beckwith, Fellbaum, Gross, & Miller, 1990) is a lexical thesaurus in the form of a database. It was first developed for English in 1990s and later for other languages. Words within WordNet are grouped in synsets, or sets of synonyms representing relations between words. The types of relations include hypernyms, hyponyms, or meronyms. Moroever, adjectives include information on related nouns and other similar words. At present WordNet contains 155,287 words organized in 117,659 synsets. A total number of word-sense pairs is 206,941. As for the Japanese version of WordNet,

[1] http://wordnet.princeton.edu/

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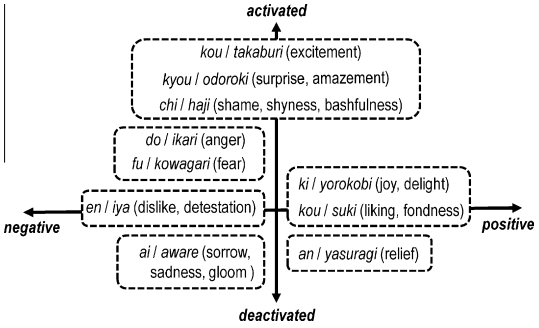


Fig. 1. Mapping of Nakamura's classification of emotions on Russell's 2D space.

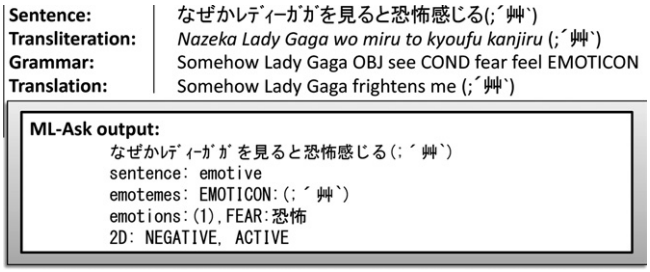


Fig. 2. Output examples for ML-Ask.

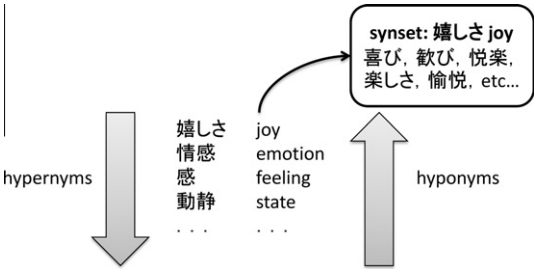


Fig. 3. Examples of synsets, hypernyms and hyponyms related to word *ureshisa* ("joy").

it was developed in 2008 and 2009. It contains 93,834 words organized in 57,238 synsets and 158,058 synset-word pairs. An example of synsets, hypernyms and hyponyms related to word *ureshisa* ("joy") are represented in Fig. 3.

Apart from the above there exist modifications of standard WordNet, such as WordNet-Affect (Strapparava & Valitutti, 2004) or Senti-WordNet (Baccianella, Esuli, & Sebastiani, 2010), which contain in particular words and synsets applicable in affect and sentiment analysis. Unfortunately, at present they are available inly for English, although there has been some research on developing their Japanese versions (Torii, Das, Bandyopadhyay, & Okumura, 2011).

4. Method for affect analysis of characters in narrative

In the task of affect analysis it is crucial to obtain not only the knowledge about the emotions that appeared in the conversation, but also about who expressed them and towards whom/about what they were expressed. This introduces to affect analysis the task of recognizing relations of particular characters with the emotions that appeared.

4.1. Method description

The method consists of three general parts. Firstly, the analyzed text is preprocessed to make the character extraction easier. Secondly, the characters represented by grammatical subject in the sentence are extracted with the use of rules based on combined morphological and syntactic information. Thirdly, affect analysis procedure is performed for sentences in which the subject was identifiable. Below we describe each part of the method. We also compared two methods for affect analysis: the first based on Japanese WordNet (Bond et al., 2009), the second based on ML-Ask system described in Section 3.2. The outline of the method is represented on Fig. 4.

4.1.1. Preprocessing

The preprocessing consists of the following steps.

1. Supplementation of subject in cases of ambiguity (if the subject is omitted),
2. supplementation of subject if the subject is a pronoun,
3. text normalization in cases of ambiguous transcription,
4. preparation of a list of characters appearing in the narrative and character naming unification.

Preparation of the list of characters and subject supplementation are aimed to facilitate the process of character determination. An example of a case covered by 1. including the disambiguation process is represented in the sentence below.

Sentence: *Okāsan wa ie ni kaettekita toki, doa ga aiteirunoni ki ga tsuita. Kowaku natte shimatta.*
Grammar[2]**:** Mother TOP home DAT return PAST moment, doors SUB open COP PAST. Afraid become UNIN PAST.
Translation: When mother came back home, the doors were open. [She] became anxious.
If no subject is present (cases of zero anaphora), like in the second sentence, a default subject is set as the first previously determined topic (TOP), usually appearing in Japanese before particle "wa", or subject (SUB), often appearing before particle "ga" (in cases where no topic appeared previously). In this case, the closest previous topic/subject ("mother") appears in the first sentence. Therefore the default subject for the second sentence is also set as "mother".

An example of a case covered by 2. including the disambiguation process is represented in the sentence below.

Sentence: *Okāsan wa ie ni kaettekita toki, doa ga aiteirunoni ki ga tsuita. Kanojo wa kowaku natte shimatta.*
Grammar: Mother TOP home DAT return PAST moment, doors SUB open COP PAST. She TOP afraid become UNIN PAST.
Translation: When mother came back home, the doors were open. She became anxious.
In cases of sentences containing multiple clauses, and in multiple sentences subjects in subsequent clauses are often replaced by pronouns (typical cases of anaphora). In such situations the subject in question is determined as the first previously determined topic (or subject if no topic was present) of the same type as the pronoun (person, animated/unanimated object). In this case, the closest previous topic/subject of the same type ("mother") appears in the second clause. Therefore the default subject for the final clause is also set as "mother".

[2] **Grammatical information in accordance to Leipzig glossing rules** (Comrie, Haspelmath, & Bickel, 2004)**. UNIN refers to** *-shimau*, **a syntactic marker of non-intentionality and undesirability of action.**

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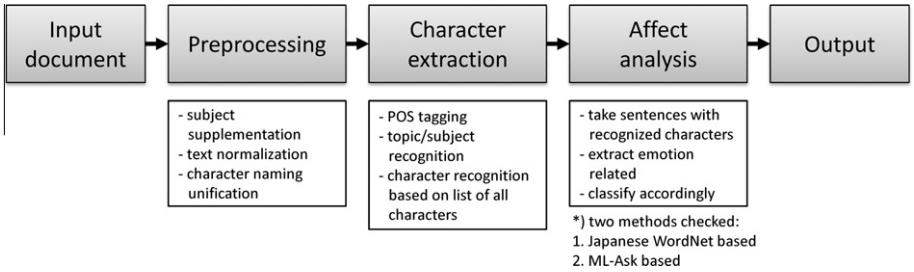


Fig. 4. Outline of the method for affect analysis of characters in narrative.

Text normalization (point 3.) is performed to facilitate the process of part-of-speech tagging used in character extraction. In Japanese language it mostly refers to normalization of transcription. Japanese is usually written with a mixture of *kanji* (chinese ideograms conveying word meaning), *hiragana* (syllabary used for grammatical information and simple words), and *katakana* (used for loan words). If most of the sentence is written in *hiragana*, the syllabograms representing words are normalized to their standard transcription (in either *kanji* or *katakana*). An example of a sentence before and after normalization is represented below.

Sentence before normalization: おかあさんはいえにかえってきた。
Sentence after normalization: お母さんは家に帰ってきた。
Sentence in romanized transcription[3]**:** *Okāsan wa ie ni kaettekita.*
Grammar: Mother TOP home DAT return PAST.
Translation: Mother came back home.

The preparation of a list of characters appearing in the narrative (point 4.) has a purely practical purpose and for the time cannot be done fully automatically. Many cases, but not all possible, are covered by the below example. Due to the possibility of a mistake we perform a manual correction in this step.

Sentence: *Okāsan wa ie ni kaettekita toki, doa ga aite ita. Onna wa kowaku natte shimatta.*
Grammar: Mother TOP home DAT return PAST moment, doors SUB open COP PAST. Woman TOP afraid become UNIN PAST.
Translation: When mother came back home, the doors were open. The woman became anxious.
In cases such as above we assume that "woman" refers to "mother".

4.1.2. Story character extraction

From the narrative preprocessed as in Section 4.1.1, sentences containing indications of characters of the narrative are selected and the characters are specified. To do this firstly the sentences are tagged with parts of speech (POS) by a Japanese POS tagger MeCab (Kudo, 2001) and parsed by a Japanese dependency parser CaboCha (Kudo & Matsumoto, 2002). The information provided by these two systems is applied similarly to the algorithm proposed by Ogi and Suganuma (2009) for the detection of topics/subjects and predicates in sentences:

(i) Consider the clause ending with a period as the clause containing the main predicate.
(ii) Consider clauses related to the predicate clause as potentially containing candidates for subject clause.
(iii) Within the clause with subject candidates find particles *wa* (topic marker) and *ga* (subject particle).
(iv) From the candidates appearing before *wa* and *ga* consider those related to the predicate clause as the subject/topic.

In the preprocessing phase all sentences are normalized and subjects are supplemented. Moreover, the list of narrative characters is prepared. A cross-reference of clauses containing topic/subject with the list of characters allows extraction of only those sentences containing a character of the narrative as the topic/subject of the sentence.

4.1.3. Affect analysis

In this section we describe two methods for affect analysis. The first one based on Japanese WordNet (baseline) and the second one based on ML-Ask.

WordNet (Baseline). The procedure for applying Japanese WordNet in affect analysis is as follows. The sentence is tagged with parts of speech and from each sentence nouns, verbs and adjectives are extracted for further analysis. All those words are separately queried in Japanese WordNet. If the word relates to emotions on any level or hypernym of hyponym, it is extracted as a candidate for emotion classification. As an example, hypernyms of word *kanashimi* ("sadness, unhappiness") are represented in Fig. 5. The word *kanashimi* is included in many synsets and thus multiple results can be obtained for this word. For example, gradual hypernyms of *kanashimi* are *kan* ("feeling"), *dōsei* ("state"), *atoribyūto* ("attribute") and *shashō* (abstract_entity). For "sorrowfulness" there is an hypernym *fuyo* ("unhappiness"), and for "gloominess" there are such hypernyms as *fukigen* ("uncheerfulness"), *atoribyūto* ("attribute") and *shashō* (abstract_entity).

In particular, in case of nouns we assume that if within synsets appear hypernyms/hyponyms like *jōkan* ("emotion") and *kan* ("feeling") the word is emotion-related. We assume that a verb is emotion-related if among its synsets appear either *kanjiru* ("[to] feel") or *kanjiru* ("[to] experience"). In case of adjectives, the computation is somewhat different. Adjectives themselves do not contain synsets like hypernyms or hyponyms. However, they contain nouns and verbs related to those adjectives. Therefore we perform an extrapolation to the synsets of those related words to check if these relate to emotions (appearance of *jōkan* (emotion), *kan* (feeling), or *kanjiru* (to feel/experience)). If they do we assume the adjectives are emotion-related as well.

ML-Ask-simple. As the second method we used the information obtained by analysis of sentences by ML-Ask. However, some modifications needed to be made to the main system. ML-Ask was designed to analyze conversations, not narratives. In the first step of ML-Ask analysis the system specifies if a sentence is emotive or non-emotive. Analysis of particular emotion types is performed only on emotive sentences. A sentence is emotive if it contains at least one *emoteme*, or a marker of emotive context. Emotemes are typical in conversations (in particular spontaneous conversations). Generally perceived narratives contain at least two main types of sentences:

[3] **Transcription to alphabet.**

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悲しさ unhappiness
⇒感 feeling
⇒動静 state
⇒アトリビュート atribute
⇒捨象 abstract_entity
悲しみ sorrow
悲しみ sorrowfulness
⇒不予 unhappiness
悲しみ gloominess
⇒不機嫌 uncheerfulness
⇒アトリビュート atribute
⇒捨象 abstract_entity

Fig. 5. Hypernyms and other words related to word *kanashimi* (“sadness”) in Japanese WordNet.

1. **descriptive sentences** for introduction of the main storyline, and
2. **dialogs** between characters of the narrative.

ML-Ask baseline can be expected to deal with the second type of sentence. However, since emotemes rarely appear in descriptive sentences, the system would not proceed to the recognition of particular emotion types. Therefore, to allow ML-Ask deal with descriptive sentences as well we excluded emotemes from the analysis and focused primarily on analysis of emotion types. However, we retained the analysis of CVS and Russell’s emotion space. Since in this version of the system we simplified the analysis, we called it **ML-Ask-simple** (later abbreviated to ML-Ask).

5. Evaluation experiment

5.1. Dataset

To evaluate the method we needed to choose a narrative. A popular kind of narratives applied in affect analysis are children stories and fairy tales (Alm et al., 2005; Mohammad, 2011). This kind of narrative usually has a clear structure and distinction between good and evil characters. Fairy tales are written in a language simple enough to be understood by young children, with comprehensible descriptions of emotions felt by the characters. Therefore we decided to evaluate the method on children stories. As the source of the dataset we used *Aozora Bunko*.[4]

Aozora Bunko (“Blue Sky Library”) is a Japanese online digital collection of freely available books (online library). It provides over ten thousand books of various genres (fiction, non-fiction) published in Japanese, for which copyrights had expired (50 years since the death of copyrights holder). In Japan, *Aozora Bunko* is sometimes compared to Project Gutenberg (Tamura, 2006).

One of the genres in *Aozora Bunko* is *Jidōsho* (“Books for children”). This category includes sub-categories, such as *Rekishi* (“History Books”), *Geijutsu-Bijutsu* (“Beautiful Arts and Crafts Books”), and *Bungaku* (“Literature”). Children stories and fairy tales are included in the *Bungaku* category. It covers 843 books for the date (March 2012). From this database we chose one fairy tale at random for the experiment.

“*The Wolf and the Seven Young Kids*” – *Overview of the Story*. The story of our choice was “*Ōkami to nana-hiki no kodomo yagi*” (“The Wolf and the Seven Young Kids”), a fairy tale by the Grimm Brothers (original title: “Der Wolf und die sieben jungen Geißlein”) from the year 1786, translated by Masao Kusuyama in 1884 and published in 1949. A representative illustration for the story is represented in Fig. 6.

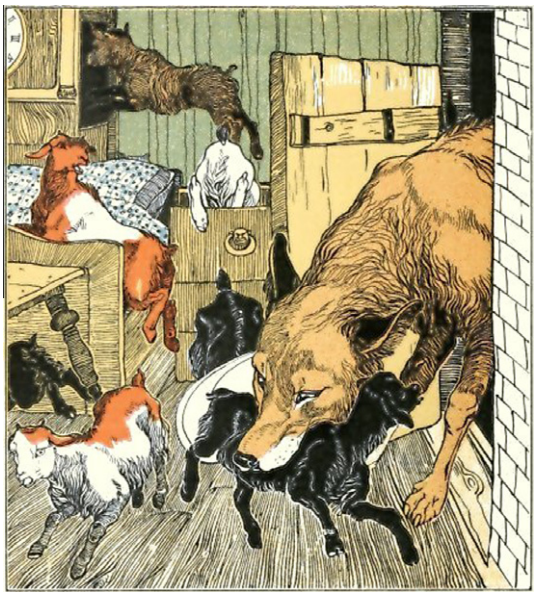


Fig. 6. “Illustration for the fairy tale “The Wolf and the Seven Young Kids” depicting the wolf attacking the kids once he is let into their home”. http://en.wikipedia.org/wiki/File:Wolf_and_7_kids.jpg, after Märchen-Sammlung von L. Bechstein. Project Gutenberg edition. <http://www.gutenberg.org/etext/23787>. Illustration by Karl Fahringer (1874–1952).

The outline of the story can be presented in the following eleven parts.

1. The Goat Mother leaves the Goat Children home alone to look after the house. Before leaving the Mother alerts the children to be on their guard for the wolf. The children reply they will take care of themselves.
2. The Wolf comes to the Goats’ house. He pretends to be the Mother by changing his voice, but the Children recognize the Wolf by the rough voice.
3. The Wolf changes his voice to sound more softly (by eating a chalk) and comes again to the Goats’ house. Again the Children recognize him by the color of his foot claws.
4. The Wolf bleaches his foot by putting some flour on it. The Children believe the Wolf and open the doors.
5. The Wolf eats all Children Goats except one. After that he falls asleep under a tree.
6. Mother comes back home. She looks for her Children in the devastated house.
7. The Mother listens to the explanation by the one Child Goat that was not eaten by the Wolf.
8. The Mother finds the Wolf, cuts his stomach with scissors and takes out her Children.
9. Mother quiets down the Children jumping of joy, and tells them to bring some stones.
10. While the Wolf is sleeping the Mother puts the stones in his stomach and sews it together. Then she flees from the scene. The Wolf wakes up, but he cannot move because of the stones.

[4] *Aozora Bunko*, http://www.aozora.gr.jp/.

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Table 3
Annotations of Emotions by 9 evaluators (gold standard, agreement ratio in the brackets), and the two systems (WordNet and ML-ask) for all parts of the story. Success cases: bold type font.

Story part	Goat Mother			Goat Children			Wolf		
	Gold standard (agreement ratio)	Wordnet	ML-Ask	Gold standard (agreement ratio)	Wordnet	ML-Ask	Gold standard (agreement ratio)	Wordnet	ML-Ask
1	Joy (66%)	**Joy**	**Fondness, relief, fear**	Joy (66%)	None	Fear	None (100%)	**None**	**None**
2	None (100%)	None	None	Anger (88%)	None	None	Anger (44%)	None	**Anger**
3	None (100%)	**None**	**None**	Anger (88%)	None	None	None (33%), Anger (33%)	**None**	**None**
4	None (100%)	**None**	**None**	Anger (33%), Joy (33%)	None	None	Anger (55%)	None	**Anger**
5	None (100%)	**None**	**None**	Sadness (88%)	None	Surprize	Joy (66%)	None	None
6	Sadness (77%)	None	None	Sadness (77%)	None	None	None (100%)	**None**	**None**
7	Sadness (100%)	None	**Sadness**	None (55%)	**None**	**None**	None (44%), Joy (44%)	**None**	**None**
8	Joy (77%)	**Joy**	**Joy**	Joy (55%)	**Joy**	None	None (77%)	**None**	Joy
9	Anger (77%)	Joy	Joy	Joy (100%)	**Joy**	None	None (88%)	**None**	**None**
10	None (66%)	**None**	**None**	None (55%)	**None**	**None**	None (66%)	**None**	**None**
11	Joy (100%)	None	**Joy**	Joy (100%)	**Joy**	**Joy**	Sadness (100%)	None	None

11. The Wolf attempts to drink water from a well, but falls inside. The Goat Mother together with the Goat Children cries out "The wolf is dead! The wolf is dead!" together they dance for joy around the well.

A story divided roughly in these 11 parts was presented to nine evaluators (all of them were graduate students). The evaluators were to read the story and annotate emotions for each of the character in each of the eleven parts of the story. The evaluators were encouraged to annotate emotions spontaneously with simple and instinctive vocabulary, like "joy", "sadness", or "anger". The annotations were then automatically classified according to the emotion classification applied in the research. In case of no emotions the evaluators were told to annotate "none" on the person/story part. This constituted the gold standard.

The story was then processed as a whole by the system including two versions of affect analysis (WordNet and ML-Ask). After the processing the extracted and analyzed characters were allocated accordingly to their appearance in all 11 parts of the story. The results of the system were then compared to the gold standard. All annotations are represented in Table 3.

5.2. Evaluation criteria

We evaluated whether the system output was equal to the annotations set by the evaluators. However, since there were some cases of disagreement about the emotions, in the evaluation we used only those types for which more than 50% of the people agreed (at least 5 people). The percentage of agreement, or agreement ratio is represented in Table 3 next to the emotion type annotated by evaluators. For some cases the agreement was 100%. In cases where not all evaluators agreed, we did not include the outliers' annotations. However, even in this setting there were three ambiguous cases. For the Goat Children in part 4 of the story, there were two most agreed emotions: anger (33%) and joy (33%). For the Wolf there were two cases of ambiguity, in part 2 of the story three people (33%) agreed about anger and three about the lack of emotions. Similarly, in part 7 four people (44%) agreed on joy and other four on the lack of emotions. In the evaluation we verified whether the ambiguity improved or impaired the analysis. We also looked at the results in two different ways. Firstly, since the goal was to annotate emotions, we assumed that cases of no emotions are irrelevant and excluded them from the evaluation. Secondly, since both, the human evaluators and the systems had the option of "no emotion", we checked the performance of the systems when the cases of no emotion were included in the evaluation. The accuracy of the systems' performance was calculated as a standard ratio of successful cases to all gold standard cases, as represented in Eq. (1).

$$Accuracy = \frac{successful\ annotations}{gold\ standard\ annotations} \quad (1)$$

Accuracies of all versions of the system are represented in Table 4. For ML-Ask we also checked two options. Firstly, when the annotated emotion types are exactly the same as those in gold standard (types-strict). Secondly, when the annotated emotion types, if not exactly the same as in gold standard, at least share the same Russell's space (types-relaxed). We also calculated statistical significance for the differences between WordNet-based method and ML-Ask.

5.3. Results

The results were as follows. For most cases the method based on ML-Ask was better than the WordNet-based method. The results were somewhat close, with 0.576 as the best result for WordNet and 0.606 as the best result for ML-Ask (difference equal to 0.03). Statistical significance of differences was confirmed for two out of three sets of results. The results were very statistically significant ($p < 0.01$) for the Goat Mother and extremely statistically significant ($p < 0.001$) for the Goat Children, but not quite statistically significant for the Wolf. However, detailed analysis of the results revealed that WordNet method was able to extract only emotions from the class of "joy", while ML-Ask was able to extract various emotion types, including "joy", "fondness", "relief", "fear", "sadness", or "anger". Therefore it can be said that ML-Ask was more balanced in affect analysis, while WordNet was more biased (towards expressions of "joy") when it comes to determination of emotion types.

Analysis with the type "no emotion" included showed better performance in both methods, which means that the system, although not always extracting emotions when it should, it rarely extracts emotions from sentences that are not emotional. As for the two options of ML-Ask evaluation, where the extracted types are considered as strict or relaxed, the relaxed method achieved the best result of all. Moreover, the results compared to the strict evaluation were statistically significant on a standard 5% level.

Detailed analysis of errors showed interesting dependencies. In scene 9, where the Mother quiets down the Children jumping of joy, and tells them to bring some stones, the Mother Goat was annotated by most of human evaluators (70%) as filled with anger. Both systems however annotated "joy". This was due to the fact that although "joy" is expressed here directly, and the mother does feel happy because the Children are safe, the dominating emotion here is anger (for Children to be quiet and for the Wolf in general). However, the information about the anger is not present directly in any form. It is inferred indirectly by the situation (context). For hu-

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Table 4
The summarized results for all kinds of evaluation criteria.

	Goat Mother			Goat Children			Wolf			Overall/average		
	WordNet	ML-Ask[b]		WordNet	ML-Ask[c]		WordNet	ML-Ask		WordNet	ML-Ask	
		Types-strict	Types-relaxed		Types-strict	Types-relaxed		Types-strict	Types-relaxed		Types-strict	Types-relaxed[a]
With non-emo	0.636	0.727	**0.818**	**0.455**	0.273	0.273	0.636	**0.727**	**0.727**	0.576	0.576	**0.606**
Only emo	0.333	0.500	0.667	0.333	0.111	0.111	0.333	0.667	0.667	0.333	0.426	0.481
Partial average	0.485	0.614	0.742	0.394	0.192	0.192	0.485	0.697	0.697	0.455	0.501	0.544
Without ambiguous sentences												
With non-emo	0.636	0.727	**0.818**	**0.500**	0.300	0.300	0.556	**0.667**	**0.667**	0.564	0.565	0.595
Only emo	0.333	0.500	0.667	0.375	0.125	0.125	0.000	0.500	0.500	0.236	0.375	0.431
Partial average	0.485	0.614	0.742	0.438	0.213	0.213	0.278	0.583	0.583	0.400	0.470	0.513

[a] $p < 0.05$.
[b] $p < 0.01$.
[c] $p < 0.001$; with comparison to baseline (WordNet).

mans this ability is inherent. The mechanism of this reasoning should be studied deeply in the future.

All annotations of particular emotion types provided by evaluators (gold standard) along with all annotations given by both systems are represented in Table 3. The results for all kinds of evaluation criteria are summarized in Table 4.

6. Conclusions and future work

In this paper we presented our research in text-based affect analysis (AA) of narratives. Affect analysis within the context of narratives is a challenging task for several reasons. Firstly, different types of sentences are used in narratives (descriptions, dialogs, etc.). Secondly, different characters become subjects of different emotional expressions in different parts of narratives. In this research we addressed in particular the problem of person/character related affect recognition in narratives. In this method firstly emotion subject is extracted from a sentence based on analysis of anaphoric expressions. Next, the affect analysis procedure estimates what kind of emotional state each character is in for each part of the narrative. We compared two methods for affect analysis. The first method was based on Japanese version of WordNet, a thesaurus consisting of words with specified relations to each other. The second method was based on ML-Ask, a system for affect analysis of Japanese language. The evaluation showed that the ML-Ask achieved significantly higher results. Although the results in general are not ideal (60.6%), they were still promising as this was the first attempt to combine two tasks: affect analysis in narratives and story characters.

We plan to improve the method in several ways. Firstly, the affect analysis system can be improved by expanding its emotion lexicon. We plan to utilize Ptaszynski et al.'s SPEC system (Ptaszynski, Rzepka, Araki, & Momouchi, 2010) to expand the number of syntactic patterns in ML-Ask. Secondly, the affect analysis procedure can be improved by applying a Web-mining technique for emotion association extraction, which showed an improvement in the past (Ptaszynski et al., 2009a).

Affect analysis of narratives is an important task crucial in language understanding. The ability to distinguish which character is in what emotional state at a given moment is especially important in this task. We showed this task is feasible to some extent for the Japanese language with available tools. Other important ones include the ability to determine why one expresses the emotion (determination of emotion object), ability to distinguish whether the expression of emotion is appropriate to the situation, and what would be the appropriate reaction. In this research we focused on the first task on the list, however, we plan to gradually address all of the above tasks to contribute to deeper understanding of emotion-related phenomena.

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References

Alm, C. O., Roth, D., & Sproat, R. (2005). Emotions from text: Machine learning for text based emotion prediction. In *Proceedings of HLT/EMNLP* (pp. 579–586).
Aman, S., & Szpakowicz, S. (2007). Identifying expressions of emotion in text. In *Proceedings of the 10th international conference on text, speech, and dialogue (TSD-2007), Plzen, Czech Republic. Lecture notes in computer science (LNCS)*. Springer Verlag.
Tamura, A. (2006). Novelists, others want copyright protection extended. The Japan Times Online.2006. September 30.
Baccianella, S., Esuli, A., & Sebastiani, F. (2010). "SentiWordNet 3.0: An enhanced lexical resource for sentiment analysis and opinion mining, LREC 2010.
Bond, F., Isahara, H., Fujita, S., Uchimoto, K., Kuribayashi, T., & Kanzaki, K. (2009). Enhancing the Japanese WordNet. In *The 7th workshop on asian language resources. In conjunction with ACL-IJCNLP 2009*.
Calix, R. A., & Knapp, Gerald M. (2011). Actor level emotion magnitude prediction in text and speech. *Multimedia Tools and Applications*, 1–14. http://dx.doi.org/10.1007/s11042-011-0909-8.
Comrie, B., Haspelmath, M., & Bickel, B. (2004). Leipzig glossing rules: Conventions for interlinear morpheme-by-morpheme glosses. Developed jointly by: Department of Linguistics of the Max Planck Institute for Evolutionary Anthropology and Department of Linguistics of the University of Leipzig. <http://www.eva.mpg.de/lingua/resources/glossing-rules.php>.
Elliott, C. (1992). The affective reasoner – A process model of emotions in a multi-agent system, PhD thesis, Northwestern University, May 1992. The Institute for the Learning Sciences, Technical Report No. 32.
Fukkoshi, H., & Sugimoto, F. (2010). A method to classify emotion in the sentences in novels. In *Proceedings of FIT2010* (pp. 227–228) [in Japanese].
Grefenstette, G., Qu, Y., Shanahan, J. G., & Evans, D. A. (2004). Coupling niche browsers and affect analysis for an opinion mining. In *Proceedings of RIAO-04* (pp. 186–194).
Kudo, T. (2001). MeCab: Yet another part-of-speech and morphological analyzer. <http://mecab.sourceforge.net/>.
Kudo, T., Matsumoto, Y. (2002). Japanese dependency analysis using cascaded chunking. In *Proceedings of the 6th conference on natural language learning 2002 (CoNLL 2002)* (pp. 63–69).
Liu, H., Lieberman, H., & Selker, T. (2003). A model of textual affect sensing using real-world knowledge. In *Proceedings of IUI 2003* (pp. 125–132).
Miller, G. A., Beckwith, R., Fellbaum, C. D., Gross, D., & Miller, K. (1990). WordNet: An online lexical database. *International Journal of Lexicography, 3*(4), 235–244.
Mohammad, S. (2011). From once upon a time to happily ever after: tracking emotions in novels and fairy tales. In *Proceedings of the ACL 2011 workshop on language technology for cultural heritage social sciences and humanities LaTeCH* (pp. 105–114).
Nakamura, A. (1993). Kanjo hyogen jiten [Dictionary of Emotive Expressions] (in Japanese), Tokyodo Publishing, Tokyo.

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Neviarouskaya, A., Prendinger, H., & Ishizuka, M. (2010). *Affect analysis model: Novel rule-based approach, to affect sensing from text*. *Natural Language Engineering* (vol. 17, no. 1). Cambridge University Press.

Ogi, H., & Suganuma, A. (2009). Detection of "Sentence of the twist" in Japanese document polishing support. In *Proceedings of IPSJ Kyushu Chapter Symposium "Hi no Kuni Joho Shinpojiumu" 2009* C-4-1, 20090314.

Ptaszynski, M., Dybala, P., Rzepka, R., Araki, K. (2009). Affecting corpora: Experiments with automatic affect annotation system – A case study of the 2channel forum. In *Proceedings of the conference of the Pacific Association for Computational Linguistics (PACLING-09)* (pp. 223–228).

Polanyi, L., & Zaenen, A. (2006). Contextual valence shifters. In J. G. Shanahan, Y. Qu, & J. Wiebe (Eds.), Computing attitude and affect in text (pp. 1–10). Dordrecht, The Netherlands: Springer Verlag.

Ptaszynski, M., Dybala, P., Shi, W., Rzepka, R., & Araki, K. (2009a). A system for affect analysis of utterances in japanese supported with Web mining. *Journal of Japan Society for Fuzzy Theory and Intelligent Informatics* [Special Issue on Kansei Retrieval, Vol. 21, No. 2 (April), pp. 30–49 (194–213)].

Ptaszynski, M., Dybala, P., Shi, W., Rzepka, R., & Araki, K. (2009b). Towards context aware emotional intelligence in machines: Computing contextual appropriateness of affective states. In *Proceedings of Twenty-first International Joint Conference on Artificial Intelligence (IJCAI-09), Pasadena, California, USA* (pp. 1469–1474).

Ptaszynski, M., Maciejewski, J., Dybala, P., Rzepka, R., & Araki, K. (2010). CAO: Fully automatic emoticon analysis system. In *Proceedings of the 24th AAAI conference on artificial intelligence (AAAI-10)* (pp. 1026–1032).

Ptaszynski, M., Rzepka, R., Araki, K., & Momouchi, Y. (2012). Automatically annotating a five-billion-word corpus of Japanese blogs for affect and sentiment analysis. In *Proceedings of the 3rd workshop on computational approaches to subjectivity and sentiment analysis (WASSA 2012), held in conjuntion with The 50th Annual Meeting of The Association for Computational Linguistics (ACL 2012) conference* (pp. 89-98), Jeju, Korea.

Ptaszynski, M., Rzepka, R., Araki, K., & Momouchi, Y. (2010). Language combinatorics: A sentence pattern extraction architecture based on combinatorial explosion. *International Journal of Computational Linguistics (IJCL), 2*(1), 24–36.

Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology, 39*(6), 1161–1178.

Shi, W., Rzepka, R., Araki, K. (2008). Emotive information discovery from user textual input using causal associations from the internet. In *FIT2008* (pp. 267–268) [In Japanese].

Strapparava, C., & Valitutti, A. (2004). "WordNet-Affect: an affective extension of WordNet. In *Proceedings of the 4th international conference on language resources and evaluation (LREC 2004)* (pp. 1083–1086).

Tokuhisa, R., Inui, K., Matsumoto, Y. (2008). Emotion classification using massive examples extracted from the web. In *Proceedings of Coling 2008* (pp. 881–888).

Torii, Y., Das, D., Bandyopadhyay, S., Okumura, M. (2011). Developing Japanese WordNet affect for analyzing emotions. In *Proceedings of the 2nd workshop on computational approaches to subjectivity and sentiment analysis (WASSA 2011), (June 2011)* (pp. 80–86).

Tsuchiya, S., Yoshimura, E., Watabe, H., & Kawaoka, T. (2007). The method of the emotion judgement based on an association mechanism. *Journal of Natural Language Processing, 14*(3), 219–238.

Wiebe, J., Wilson, T., & Cardie, C. (2005). Annotating expressions of opinions and emotions in language. *Language Resources and Evaluation, 39*(2-3), 165–210.

Wilson, T., & Wiebe, J. (2005). Annotating attributions and private states. In *Proceedings of the ACL workshop on frontiers in corpus annotation II* (pp. 53–60).

Zhang, L. (2010). Exploitation of contextual affect-sensing and dynamic relationship interpretation. In *ACM Computers in Entertainment* (Vol. 8, Issue 3).